



Course project

Due Dates:

Proposal (Due: 20 -4 - 2026)

Project Reports (Final Draft due on 2 -5 - 2026)

Presentation (3 -5 - 2024) and 5-5-2026

Submission Method:

Using BB (link will be supplied later)

Rules:

- The project can be developed personal or with group
- Similarities should not exceed 15%
- Report and presentation are needed

Presentation method

In Presence or Online (will be decided later)

Project Description

The default project is to do some original research in a group of three students. For example:

- 1- Implementation of chatting system
- 2- Making animation movies (not less than 10 minutes)
- 3- Implementation of any compression algorithm not covered in the lab
- 4- Implementation of JPEG algorithm
- 5- Implementation of JPEG2000 algorithm

- 6- Implementation of MPEG4 algorithm
- 7- Voice compression
- 8- File format converter
- 9- MP3 format coding
- 10- AAC (Advanced Audio Code) audio compression
- 11- MCC simulation
- 12- Design a simple multimedia game
- 13- Simulation of Multimedia traffic using NS2

Such a project would involve using system manuals, architecture descriptions, and research papers to present a clear and coherent of the project.

You will be graded on (1) how well you define your problem, (2) survey previous work, (3) design, coding and conducting simulation, and (4) present your results.

Report=10%, Presentation=10%, Implementation 10%

Proposal (Due: 20 -4 - 2026)

Proposals should be about one to two pages long. All proposals should include: a description of your topic, a statement of why you think the topic is interesting or important, a proposed schedule for your work, and references to at least three papers you have obtained.

You may use the Digital Libraries on www and Readings in Computer Architecture Web Component.

Project Reports (Due 2-5-26)

Reports should consist of an abstract, body and optional appendices. The abstract should summarize the contributions of the report in one or two paragraphs. The length of the body should be the equivalent of 3-5 pages at double-spaced 10-point. Additional supporting material of any length can be put in appendices.

Presentation (3 -5 – 2026) and (5-5-2026)

The presentation will be 10-20 minutes in length depending on the number of groups. All group members should deliver part of the talk. The talk should give highlights of the final report, including the problem, motivation, results, conclusions, and possible future work. Time limits will be enforced to let everyone present. We will meet with each group to review your proposed talk in the week prior to the presentations.

References:

- **Loco project** (<http://www.hpl.hp.com/loco/>)
- **NETCICATS project** (<http://www.eecis.udel.edu/~iren/netcicats.html>)
- **The MPEG homepage:** <http://drogo.cselt.stet.it/ufv/leonardo/mpeg/>
- **Multimedia Information Systems**
by [V. S. Subrahmanian](#) (Editor), [S. K. Tripathi](#) (Editor)
- Java Media API
- <http://www.javasoft.com/products/java-media/jmf/index.html>
- Java Advanced Imaging API
<http://www.javasoft.com/products/java-media/jai/index.html>